

4(a)(i)	<u>565 nm</u> ;	1
4(a)(ii)	1 phycoerythrin does not absorb light above 590–600 (nm) ; A suitable value for absorption that answer the question 2 (but) photosynthesis still takes place (above 590–600 nm) ; A suitable value for absorption that answer the question	2

Question	Answer	Marks
4(a)(iii)	any two from: 1 <u>accessory</u> pigment ; 2 absorbs <u>light energy</u> (up to 595 nm) ; R trap / capture 3 (and) passes it to, primary pigment / reaction centre / chlorophyll a ;	2
4(b)(i)	<u>chromatography</u> ;	1
4(b)(ii)	any three from: 1 calculate R_f value(s) ; 2 compare R_f values of pigments (in red algae and green plant) ; 3 find pigment present (on chromatogram) from red algae but absent from green plant or R_f value of phycoerythrin will not be the same as R_f values of pigments in green leaves ; 4 identify pigment using reference values (to confirm it is phycoerythrin) ;	3